2.3. MATRIX MULTIPLICATION

Suppose $T:\mathbb{R}^k \to \mathbb{R}^n$ and $S:\mathbb{R}^n \to \mathbb{R}^m$ are two transformations. The composite of S and T is the map $S\circ T:\mathbb{R}^k \to \mathbb{R}^m$ given by

$$(S\circ T)(\vec{x}) = S(T(\vec{x}))$$

Recall that $S\circ T$ and $T\circ S$ are not the same in general (in fact, it could be the case that one of the composites exists while the other one does not).

Now, let's consider an $m\times n$ matrix A and an $n\times p$ matrix B. These define matrix transformations $T_A:\mathbb{R}^n \to \mathbb{R}^m$ and $T_B:\mathbb{R}^p \to \mathbb{R}^n$. Is $T_A\circ T_B$ a matrix transformation? If $\vec{x}$ is in $\mathbb{R}^p$, we have

$$\begin{aligned}(T_A\circ T_B)(\vec{x}) &= T_A(T_B(\vec{x})) = A(B\vec{x})\\ &= A(x_1\vec{b}_1 + \dots + x_p\vec{b}_p) \quad (\text{Where } B=[\vec{b}_1 \cdots \vec{b}_p])\\ &= x_1(A\vec{b}_1) + \dots + x_p(A\vec{b}_p)\\ &= [A\vec{b}_1 \cdots A\vec{b}_p]\vec{x}\end{aligned}$$

Thus, $T_A\circ T_B$ is the matrix transformation induced by the $m\times p$ matrix $[A\vec{b}_1 \cdots A\vec{b}_p]$. The latter matrix will be called AB:

DEF: Given an $m\times n$ matrix A and an $n\times p$ matrix B, we define their product AB as the $m\times p$ matrix whose decomposition into columns is

$$AB = [A\vec{b}_1\ A\vec{b}_2 \cdots A\vec{b}_p]$$

SOME REMARKS:

- The (i,j)-entry of AB is $a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$.
- If the product AB exists (i.e. if the number of columns of A and the number of rows of B are the same), we say that A and B are compatible for multiplication.
- If A is a square matrix, we can take its powers: $A^2 = AA$, $A^3 = A^2A = AAA$, and so on.
- By design, $T_A\circ T_B = T_{AB}$ (as a consequence, the composition of matrix transformations is also a matrix transformation).

EXAMPLES:

$$\begin{bmatrix} 8 & 3 & 4 \\ -1 & 2 & 5 \end{bmatrix}\begin{bmatrix} 6 & -2 \\ 1 & 7 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 51 & 33 \\ -4 & 51 \end{bmatrix} \qquad \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ 5 & -1 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 8 & 10 \\ 12 & 6 \\ 20 & 4 \end{bmatrix}$$

WARNING! If A and B are matrices, AB and BA can be different!

EXAMPLE:

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

If $AB = BA$, we say that A and B commute.

EXAMPLE: Consider any $m \times n$ matrix A. Then $AI_n = I_m A = A$.

SOL: We have

$$AI_n = A[\vec{e}_1\ \vec{e}_2 \cdots \vec{e}_n] = [A\vec{e}_1\ A\vec{e}_2 \cdots A\vec{e}_n] = [\vec{a}_1\ \vec{a}_2 \cdots \vec{a}_n] = A$$ (where $\vec{a}_j$ is the j-th column of A for each j).

Similarly:

$$I_m A = I_m[\vec{a}_1\ \vec{a}_2 \cdots \vec{a}_n] = [I_m\vec{a}_1\ I_m\vec{a}_2 \cdots I_m\vec{a}_n] = [\vec{a}_1\ \vec{a}_2 \cdots \vec{a}_n] = A.$$

THEOREM: Let a be a scalar and A, B, C matrices such that the products below are defined. Then:

1) $IA = AI = A$
2) $A(BC) = (AB)C$ (Associative law)
3) $A(B+C) = AB + AC$ (Distributive law 1)
4) $(B+C)A = BA + CA$ (Distributive law 2)
5) $a(AB) = (aA)B = a(AB)$
6) $(AB)^T = B^T A^T$

PROOF: Most of the equations above are proved in the same way as their counterparts from Section 2.2, so we will only prove 6).

Suppose $A = [a_{ij}]$ is an $m \times n$ matrix and B is an $n \times p$ matrix. Then $A^T = [a_{ji}]$ and $B^T = [b_{ji}]$. Thus, the (i,j)-entry of $B^T A^T$ is

$$B^T(i,1)A^T(1,j) + B^T(i,2)A^T(2,j) + \cdots + B^T(i,n)A^T(n,j) = b_{1i}a_{j1} + b_{2i}a_{j2} + \cdots + b_{ni}a_{jn}$$
$$= a_{j1}b_{1i} + a_{j2}b_{2i} + \cdots + a_{jn}b_{ni} = AB(j,i) = AB^T(i,j) \qquad //$$

Example: Suppose A, B, C are matrices of the same size. Simplify the expression

$$A(BC-CB)-(BC-CB)A+B(CA-AC)-(CA-AC)B+C(AB-BA)-(AB-BA)C$$

Sol

$$A(BC-CB)-(BC-CB)A+B(CA-AC)-(CA-AC)B+C(AB-BA)-(AB-BA)C$$
$$=ABC-ACB-BCA+CBA+BCA-BAC-CAB+ACB+CAB-CBA-ABC+BAC$$
$$=0 \quad \text{(because all of the terms cancel out)}$$

Example: Let $A=\begin{bmatrix}1&0\\2&1\end{bmatrix}$. Find a matrix B that satisfies $AB=I_2$. Is $BA=I_2$?

Sol

Write $B=\begin{bmatrix}a&b\\c&d\end{bmatrix}$. Then $AB=\begin{bmatrix}a&b\\2a+c&2b+d\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}$. Thus, this matrix equation becomes the system

- $a=1$ · $2a+c=0 \Rightarrow 2+c=0 \Rightarrow c=-2$
- $b=0$ · $2b+d=1 \Rightarrow d=1$

So $\boxed{B=\begin{bmatrix}1&0\\-2&1\end{bmatrix}}$

We now compute $BA=\begin{bmatrix}1&0\\-2&1\end{bmatrix}\begin{bmatrix}1&0\\2&1\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I_2$!

As we can see, B behaves as a "multiplication inverse" of A (more on this in the next section).

A word on block multiplication

In the same way that we have decomposed a matrix into its columns, sometimes it is useful to decompose a matrix into blocks. For instance:

$$A=\left[\begin{array}{cc|cc}1&0&0&0\\0&1&0&0\\\hline 2&4&1&6\\3&0&7&8\\1&1&0&0\end{array}\right]=\begin{bmatrix}I_2&0\\P&Q\end{bmatrix}$$

(We view A as a 2×2 matrix whose entries are also matrices)

$\uparrow P \qquad \uparrow Q$

It turns out that, if A and B are partitioned compatibly into blocks, then AB can be computed using the blocks as entries

Example: $\begin{bmatrix}A&B\\0&C\end{bmatrix}\begin{bmatrix}A'&B'\\0&C'\end{bmatrix}=\begin{bmatrix}AA'&AB'+BC'\\0&CC'\end{bmatrix}$ if the matrix products on the right are defined.